ANALYSE DE LA DYNAMIQUE D'APPRENTISSAGE DES RÉSEAUX DE NEURONES PROFONDS À TRAVERS LE PRISME DU NOYAU DE TANGENTE NEURONALE

Contributions à la compréhension des corrections à largeur finie et des régimes d'apprentissage

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TABLE DES MATIÈRES

RÉSUMÉ

Ce rapport présente les résultats d'un stage de recherche axé sur l'étude de la dynamique d'apprentissage des réseaux de neurones profonds. Nous explorons les mécanismes sous-jacents à travers le formalisme du Noyau de Tangente Neuronale (NTK), en portant une attention particulière aux corrections à largeur finie. Nos travaux s'articulent autour de la décomposition du NTK via l'entraînement de Sobolev, l'analyse de son espace à noyau reproduisant (RKHS) et l'étude des valeurs propres extrêmes. Nous présentons des bornes théoriques pour la plus petite valeur propre du NTK en nous appuyant sur des techniques de matrices aléatoires (RMT). Ces résultats sont ensuite appliqués à l'analyse de différentes architectures, notamment les réseaux de neurones à propagation avant (FCNN), les réseaux à mémoire matricielle (MMNN) et les transformeurs. Nous discutons également des implications de nos découvertes pour les applications en calcul scientifique (SciML) et proposons des pistes pour de futures recherches.

EXECUTIVE SUMMARY

(Executive summary placeholder)

INTRODUCTION

L'étude de la dynamique d'apprentissage dans les réseaux de neurones profonds constitue un domaine de recherche fondamental. Le régime du Noyau de Tangente Neuronale (NTK) a fourni un cadre puissant pour analyser le comportement des réseaux infiniment larges, où la dynamique d'entraînement se linéarise. Cependant, les réseaux pratiques ont une largeur finie, ce qui introduit des corrections complexes et un comportement d'apprentissage des caractéristiques (feature learning). Ce rapport vise à approfondir notre compréhension de ces phénomènes.

Nous commençons par une analyse théorique du NTK, en présentant sa décomposition basée sur l'entraînement de Sobolev et en caractérisant son RKHS. Une partie centrale de nos travaux est consacrée à l'étude du spectre du NTK, où nous cherchons à borner ses plus petites valeurs propres à l'aide d'outils issus de la théorie des matrices aléatoires. Ces résultats théoriques sont ensuite mis en perspective à travers l'analyse de plusieurs architectures : FCNN, MMNN et transformeurs. Pour les FCNN, nous étudions les corrections à largeur finie et le scaling des paramètres. Pour les MMNN et les transformeurs, nous discutons des défis spécifiques liés à l'optimisation et des intuitions sur leurs performances.

Enfin, nous explorons les applications de nos travaux, notamment en SciML, et discutons de l'alignement entre les hypothèses théoriques et les observations empiriques. Nous abordons également des perspectives inspirées de la physique statistique pour interpréter la convergence des modèles vers l'équilibre.

NOTATIONS

$\_\!/$: La fonction d'activation ReLU (Rectified Linear Unit), définie par $\_\!/(x) = \max(0, x)$.
$\mathbb{E}[\cdot]$: Espérance math ématique.
$\mathrm{Var}(\cdot)$: Variance d'une variable aléatoire.
$\mathbb{1}\{\cdot\}$: Fonction indicatrice, valant 1 si la condition est vraie, 0 sinon.
dx : Élément différentiel utilisé dans les intégrales.

1
CADRE THÉORIQUE DU NOYAU DE TANGENTE NEURONALE

1.1 Entraînement de Sobolev et Décomposition du NTK

(Présentation des théorèmes sur la décomposition du NTK via le Sobolev training. Description du RKHS pour le noyau de Sobolev et le noyau de Laplace.)

1.2 Analyse Spectrale du NTK et Bornes sur les Valeurs Propres

(Discussion sur la plus grande valeur propre du NTK. Présentation des théorèmes de Terjek et du plan de preuve pour obtenir une borne sur la plus petite valeur propre en utilisant la RMT.)

2
CORRECTIONS À LARGEUR FINIE ET RÉGIMES D'APPRENTISSAGE

2.1 Développement Théorique pour les FCNN

(Présentation des théorèmes et formules pour les corrections à largeur finie dans les FCNN. Analyse des lois de scaling du reste et discussion sur les termes d'erreur, en lien avec les ensembles de matrices aléatoires QUE GOE.)

2.2 Approches Alternatives et Concentration

(Explication de l'approche alternative pour les FCNN via les réseaux paraboliques de Terjek, permettant de se placer en régime de feature learning tout en conservant des propriétés du NTK. Discussion sur les bornes de concentration.)

3
ANALYSE D'ARCHITECTURES SPÉCIFIQUES

3.1 Réseaux à Mémoire Matricielle (MMNN) et Transformeurs

(Description des techniques de préconditionnement du NTK pour ces architectures. Discussion sur les régimes "feature" et "kernel", et sur les défis liés à l'optimisation. Présentation d'une conjecture sur la stabilité autour des minima globaux.)

4
MMNN NTK for 2 hidden layers

4.1 recursive

For a 2 hidden layer MMNN, we can write the recursive formulation of the NTK kernel. Let's recall the general recursive formula for layer ℓ :

suppose that you have a low rank hidden layer, with a width r

so the architecture is :

$$\begin{array}{ccccccccc} (x,y) & \xrightarrow{\sigma W_1} & h_1 & \xrightarrow{A_1 \cdot + c_1} & r & \xrightarrow{\sigma W_2} & h_2 & \xrightarrow{A_2 \cdot + c_2} & (x_2, y_2) \\ & & N & & (x_1, y_1) \text{ in } \mathbb{R}^r & & N & & \end{array}$$

And here N grow to infinity, and r is fixed.

Then the NTK random kernel is :

$$\Theta^{(\ell)}(x, x') = \Theta^{(\ell-1)}(x, x') \cdot \left(\sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x')\right) + \Sigma^{(\ell)}(x, x') + 1 \tag{1}$$

For $\ell = 2$, starting from $\Theta^{(0)} = 0$, we have :

$$\begin{aligned} \Theta^{(1)} &= \mathbb{E}_{w,b}\left[\sigma(w^\top x + \beta b)\sigma(w^\top x' + \beta b)\right] + 1 \\ \Theta^{(2)} &= \Theta^{(1)} \cdot \left(\sigma_A^2 \mathbb{E}_{w,b}\left[\dot{\sigma}(w^\top h^{(1)}(x) + \beta b)\dot{\sigma}(w^\top h^{(1)}(x') + \beta b) w^\top w\right]\right) \\ &\quad + \mathbb{E}_{w,b}\left[\sigma(w^\top h^{(1)}(x) + \beta b)\sigma(w^\top h^{(1)}(x') + \beta b)\right] + 1 \end{aligned}$$

Here, $\Theta^{(1)}$ is a deterministic NNGP kernel (Gaussian Process) since it depends only on the input x, while $\Theta^{(2)}$ is a random field (Deep Gaussian Process) that depends on the realization of the first layer's output $h^{(1)}$. When $\beta = 0$, these expectations simplify by removing the bias terms b. This non-Gaussian nature of the kernels for $\ell > 1$ is a key distinguishing feature of MMNNs compared to standard neural networks.

When $\beta = 0$, the equations simplify to :

Main focus : NTK kernels with β

$$\begin{aligned}\Theta^{(1)} &= \mathbb{E}_w\left[\sigma(w^\top x)\sigma(w^\top x')\right] + 1 \\ \Theta^{(2)} &= \Theta^{(1)} \cdot \left(\sigma_A^2 \mathbb{E}_w\left[\dot\sigma(w^\top h^{(1)}(x))\dot\sigma(w^\top h^{(1)}(x'))w^\top w\right]\right) \\ &\quad + \mathbb{E}_w\left[\sigma(w^\top h^{(1)}(x))\sigma(w^\top h^{(1)}(x'))\right] + 1\end{aligned}$$

Note that for stable signal propagation at the Edge of Chaos (EOC), we require for 1 rank hidden layer :

$$\sigma_A^2 \cdot \mathrm{Tr}(\Sigma_w) = 2$$

Then for a rank r

$$\sigma_A^2 \cdot \mathrm{Tr}(\Sigma_w) = \frac{2}{r}$$

And we assume at least isotropic gaussian for the weights without loss of generality and to ensure fast computations.

This condition ensures the signal neither vanishes nor explodes as it propagates through the network layers. This also means we need to normalize by $\sqrt{r}$ the output of the first MMNN layer. We call x_1 and y_1 the outputs of the first layer unnormalized.

We recall also that $\Sigma^{(1)}$ is the arccosine kernel, which has a closed form in function of the correlation ρ between x and x' :

$$\Sigma^{(1)}(\rho) = \frac{1}{\pi}\left(\sqrt{1-\rho^2} + \rho(1-\arccos(\rho))\right)$$

We recall also that $\dot\Theta^{(1)}(\rho) = \mathbb{E}_w\left[w^\top w \dot\sigma(w^\top x)\dot\sigma(w^\top x')\right]$. We can compute this exactly when w is isotropic gaussian with variance 1 :

In fact a simple computation takes the expectation of the product of the two derivatives of the ReLU, by independence of $\|w\|^2$ and $\mathbb{1}_{w^\top x>0} \cdot \mathbb{1}_{w^\top x'>0}$:

$$\begin{aligned}\dot\Theta^{(1)}(\rho) &= \mathbb{E}_w\left[w^\top w \dot\sigma(w^\top x)\dot\sigma(w^\top x')\right] \\ &= \mathbb{E}_w\left[w^\top w \cdot \mathbb{1}_{w^\top x>0} \cdot \mathbb{1}_{w^\top x'>0}\right] \\ &= \mathbb{E}\left[\|w\|^2\right] \cdot \mathbb{P}(w^\top x > 0, w^\top x' > 0) \\ &= \mathrm{Tr}(\Sigma_w)\left(\frac{1}{2} - \frac{\arccos(\rho)}{2\pi}\right)\end{aligned}$$

It is just a reformulation of the fact that we integrate over a cone of angle $\arccos(\rho)$ a radial function.

We then write $\Theta^{(2)}$ as a function of ρ and ρ_1 :

$$\boxed{\begin{aligned}&\Theta^{(2)}(\rho,\rho_1) = \Theta^{(1)}(\rho) \cdot \sigma_A^2\,\mathrm{Tr}(\Sigma_w)\left(\frac{1}{2} - \frac{\arccos(\rho)}{2\pi}\right) + \frac{1}{r}\cdot\Sigma^{(1)}(\rho_1)\|x_1\|\|y_1\| + 1 \\ &\text{with}\quad \Sigma^{(1)}(\rho) = \frac{1}{\pi}\left(\sqrt{1-\rho^2} + \rho(1-\arccos(\rho))\right)\end{aligned}}$$

where ρ_1 is the correlation between the outputs of the first layer, that is a random variable and ρ is the correlation between the inputs.

Note that this multiplicative value is between 0 and 1, that is we can compute a lot of stuff after concatenating layers and getting some exponential scaling laws. We emphasize the fact we multiply by $\|x_1\|\|y_1\|$ which is a random variable, but as we will see, it is independent of the correlation ρ_1.

4.2 Expression for random variables ρ_1, $\|x_1\|^2$, $\|y_1\|^2$

The output of the 1st layer is a gaussian process of dimension r, where each coordinate is a gaussian process with covariance matrix :

$$\Sigma_i^{(1)} = \mathbb{E}_w \left[\sigma(w^\top x)\sigma(w^\top x')\right]$$

and the coordinates are ρ-correlated. If we stack horizontally the outputs x_1 and y_1 in a vector of dimension $2r$, the full $2r \times 2r$ covariance matrix is therefore :

$$\Sigma^{(1)} = \begin{pmatrix} I & \rho I \\ \rho I & I \end{pmatrix}$$

It's a kronecker product of r matrices I and 2d ρ correlated gaussian variables.

it happens, by a magic, that we can compute exactly the joint distribution of the random variable $(\rho_1, \|x_1\|^2, \|y_1\|^2)$:

$$\rho_1 = \frac{\sum_{i=1}^r x_i y_i}{\sqrt{\sum_{i=1}^r x_i^2}\sqrt{\sum_{i=1}^r y_i^2}}$$

The squared norms x_1^2 and y_1^2 follow a bivariate Kibble distribution and their cosine distance follows an independent Fisher like distribution. The Kibble distribution is defined for $u, v \geq 0$ and $\rho \in [-1, 1]$, while the Fisher distribution is defined for $\theta \in [-1, 1]$ and $\rho \in [-1, 1]$ with $r > 2$.

$$\begin{aligned}
(\|x_1\|^2, \|y_1\|^2) &\sim \text{Kibble}(r, \rho) \\
&= \frac{(uv)^{\frac{r/2-1}{2}} e^{-\frac{u+v}{2(1-\rho^2)}}}{\Gamma(r/2)(2(1-\rho^2))^{r/2+1}\rho^{\frac{r/2-1}{2}}} \cdot I_{r/2-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right) \\
\rho_1 &\sim \text{Fisher}(\rho, r) \\
&= \frac{(r-2)\Gamma(r-1)(1-\rho^2)^{\frac{r-1}{2}}(1-\theta^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1-\rho\theta)^{r-\frac{3}{2}}} \times {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; \frac{1+\rho\theta}{2}\right)
\end{aligned}$$

where $u = \|x_1\|^2$, $v = \|y_1\|^2$, $\theta = \cos(x_1, y_1)$, and I_α is the modified Bessel function of the first kind.

and the correlation of x,y :

where x_{1i} and y_{1i} are ρ-correlated gaussian variables.

We ensure that the reader has understood that the correlation of x,y is independent of the squared norms of x,y !

4.2.1 • Final result

We can then state the final result : For 2 entries x,y in $\mathbb{R}^d$ and a low rank r hidden layer, the NTK for a 2 hidden layer MMNN is :

$$\boxed{\begin{aligned}
&\Theta^{(2)}(x,y) = \Theta^{(1)}(\rho) \cdot \left(1 - \frac{\arccos(\rho_1)}{\pi}\right)\frac{1}{r} + \frac{1}{r}\Sigma^{(1)}(\rho_1)\|x_1\|\|y_1\| + 1 \\
&\text{with} \quad \Sigma^{(1)}(\rho) = \frac{1}{\pi}\left(\sqrt{1-\rho^2} + \rho(1 - \arccos(\rho))\right)
\end{aligned}}$$

where the random variables follow :
- $\rho_1 \sim \text{Fisher}(\rho, r)$ is independent of the norms
- $(\|x_1\|^2, \|y_1\|^2) \sim \text{Kibble}(r, \rho)$ are chi-squared variables with r degrees of freedom from correlated gaussian variables.

4.3 Intuitions sur les Performances

(Discussion finale basée sur les résultats théoriques et expérimentaux pour expliquer les performances observées pour les MMNN, les transformeurs et les ResNets.)

5 Résultats expérimentaux et applications

5.1 Validation des Lois de Scaling et des Corrections

(Présentation des expériences numériques validant les lois de scaling pour les corrections à largeur finie dans les FCNN.)

5.2 Analyse Hessienne et Dynamique

(Présentation des expériences sur l'approche hessienne du NTK et l'étude des termes de type Lyapunov.)

5.3 Applications en SciML et Discussion

(Discussion sur l'application des résultats en Scientific Machine Learning. Vérification des hypothèses théoriques sur les grands modèles et le scaling des paramètres. Lien avec la physique statistique et la convergence vers l'équilibre thermique.)

ANNEXES

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Figure 1 – Aperçu de l'interface mise à disposition en source ouverte pour l'extraction de connaissances

A
FISHER, KIBBLE DISTRIBUTIONS AND THE FAKE "CURSE OF DIMENSIONALITY"

A.1 Distribution of the Correlation Coefficient r

Fisher derived the exact law of r in 1915, representing the cosine between Gaussian vectors of dimension r with correlation ρ :

$$p(r|\rho, r) = \frac{(r-2)\Gamma(r-1)(1-\rho^2)^{\frac{r-1}{2}}(1-r^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1-\rho r)^{r-\frac{3}{2}}} \times {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho r}{2}\right)$$

Where Γ is the gamma function and ${}_2F_1$ is the hypergeometric function.

For large r, Fisher's z-transform gives $z = \text{arctanh}(r) \sim \mathcal{N}\left(\text{arctanh}(\rho) + \frac{\rho}{2(r-1)}, \frac{1}{r-3}\right)$. The variance of r decreases as $O(1/r)$. When $\rho \sim O(1/\sqrt{r})$, geometric noise masks correlation.

A.2 The Kibble Distribution

The joint law of correlated chi-squared variables $U = \|X\|^2$ and $V = \|Y\|^2$ follows Kibble's 1941 distribution :

$$f(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} e^{-\frac{u+v}{2(1-\rho^2)}}}{\Gamma(r/2)(2(1-\rho^2))^{r/2+1}\rho^{\frac{r/2-1}{2}}} \cdot I_{r/2-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right)$$

Where Γ is the gamma function and I_α is the modified Bessel function of the first kind.

Mean $\mathbb{E}[U] = r$ and variance $\text{Var}(U) = 2r$ grow linearly. U/r converges to $\mathcal{N}(1, 2/r)$ with polynomial concentration.

B
PRICE'S THEOREM AND DERIVATIONS

Price's theorem was a central tool in our analysis.

- **Statement of price's theorem**

Suppose that X_1 and X_2 forms a gaussian vector with joint distribution $f_{X_1,X_2}(x_1, x_2)$. Then, for any function $g(x_1, x_2)$, we have :

$$\mathbb{E}[g(X_1, X_2)] = \int_0^\rho \mathbb{E}\left[\frac{\partial^2 g}{\partial x_1 \partial x_2}(X_1, X_2)\right] d\rho + \text{Constant}$$

where the constant is the value of the expectation when $\rho = 0$.

- ## Calculation of I_{12} and I_{21} (First-Order Moments)

For I_{12}, we need to calculate :

$$I_{12} = \int_a^\infty \int_b^\infty z_2 \phi_2(z_1, z_2; \rho)\, dz_2\, dz_1$$

Step 1 : Inner integral We first focus on the inner integral with respect to z_2. The exact formula for the conditional density is :

$$\phi_2(x, y; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(y) \phi_1 \left(\frac{x - \rho y}{\sqrt{1-\rho^2}} \right)$$

Applying this formula with $x = z_1$ and $y = z_2$:

$$\phi_2(z_1, z_2; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(z_2) \phi_1 \left(\frac{z_1 - \rho z_2}{\sqrt{1-\rho^2}} \right)$$

The inner integral becomes :

$$\int_b^\infty z_2 \frac{1}{\sqrt{1-\rho^2}} \phi_1(z_2) \phi_1 \left(\frac{z_1 - \rho z_2}{\sqrt{1-\rho^2}} \right) dz_2$$

Step 2 : Change of variable Let $u = \frac{z_1 - \rho z_2}{\sqrt{1-\rho^2}}$. Then :

$$z_2 = \frac{z_1 - u\sqrt{1-\rho^2}}{\rho}$$
$$dz_2 = -\frac{\sqrt{1-\rho^2}}{\rho}\, du$$

The lower bound becomes $u_{max} = \frac{z_1 - \rho b}{\sqrt{1-\rho^2}}$. The integral transforms into :

$$\frac{1}{\sqrt{1-\rho^2}} \int_{-\infty}^{\frac{z_1 - \rho b}{\sqrt{1-\rho^2}}} \left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho} \right) \phi_1 \left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho} \right) \phi_1(u) \left(-\frac{\sqrt{1-\rho^2}}{\rho} \right) du$$

Step 3 : Simplification Grouping terms and using properties of ϕ_1 :

$$-\frac{1}{\rho^2} \int_{-\infty}^{\frac{z_1 - \rho b}{\sqrt{1-\rho^2}}} (z_1 - u\sqrt{1-\rho^2}) \phi_1(u) \phi_1 \left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho} \right) du$$

Step 4 : Using the conditional density formula The exact formula is :

$$\phi_2(x, y; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(x) \phi_1 \left(\frac{y - \rho x}{\sqrt{1-\rho^2}} \right)$$

Applying it correctly to our case with $x = z_1$ and $y = z_2$:

$$I_{12} = \phi_1(b) \Phi_1 \left(\frac{a + \rho b}{\sqrt{1-\rho^2}} \right) + \rho \phi_1(a) \Phi_1 \left(-\frac{b + \rho a}{\sqrt{1-\rho^2}} \right)$$

This final formula for I_{12} is one of the key terms in calculating the expectation of the product of ReLUs.

- **Calculation of I_{22} (Cross-Moment)**

The calculation of I_{22} is the most complex part. The final result is :

$$I_{22} = \int_0^\rho I_{11} + \phi_2(a, b; \rho)$$

The proof of this identity was a central point of the discussion. The most rigorous method consists of showing the equality for $\rho = 0$, and then the equality of the derivatives with respect to ρ for both sides of the equation.

- **Final Formula**

By assembling all the terms, we obtain the final algebraic formula :

$$\mathbb{E}[\dots] = (\mu_1\mu_2 + \rho\sigma_1\sigma_2)I_{11} + \mu_1\sigma_2 I_{12} + \mu_2\sigma_1 I_{21} + \sigma_1\sigma_2\phi_2(a, b; \rho)$$

B.1 Computing ReLU Product Expectations

When analyzing the Neural Tangent Kernel (NTK) of a single-layer network, we need to compute expectations of the form :

$$\mathbb{E}[\mathrm{ReLU}(X_1)\mathrm{ReLU}(X_2)]$$

where (X_1, X_2) follows a bivariate normal distribution with parameters $(\mu_1, \sigma_1, \mu_2, \sigma_2, \rho)$.

For deterministic input vectors x_1 and x_2, we define the following quantities :

— $\sigma_1 = \|x_1\|$ (where $\|\cdot\|$ denotes Euclidean norm)
— $\sigma_2 = \|x_2\|$
— $\rho = \cos(x_1, x_2) = \frac{\langle x_1, x_2 \rangle}{\|x_1\|\|x_2\|}$

The network outputs follow a Gaussian process characterized by the covariance matrix :

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

$\sigma_1^2 \rho\sigma_1\sigma_2$
$\rho\sigma_1\sigma_2\sigma_2^2$

B.2 Decomposition

We recall the expression of mu The approach is to standardize the variables ($Z_i = (X_i - \mu_i)/\sigma_i$) and decompose the integral into four terms :

$$\mathbb{E}[\dots] = \sigma_1\sigma_2 I_{22} + \mu_1\sigma_2 I_{12} + \mu_2\sigma_1 I_{21} + \mu_1\mu_2 I_{11}$$

where the integrals I_{ij} are defined over the domain $z_1 > a$ and $z_2 > b$, with $a = -\mu_1/\sigma_1$ and $b = -\mu_2/\sigma_2$.

— $I_{11} = \int_a^\infty \int_b^\infty \phi_2(z_1, z_2; \rho)\, dz_2\, dz_1$
— $I_{21} = \int_a^\infty \int_b^\infty z_1\phi_2(z_1, z_2; \rho)\, dz_2\, dz_1$
— $I_{12} = \int_a^\infty \int_b^\infty z_2\phi_2(z_1, z_2; \rho)\, dz_2\, dz_1$
— $I_{22} = \int_a^\infty \int_b^\infty z_1 z_2\phi_2(z_1, z_2; \rho)\, dz_2\, dz_1$

B.3 Calculation of the Components

• Calculation of I_{11} (Probability)

I_{11} is the probability $P(Z_1 > a, Z_2 > b)$. It is calculated via the cumulative distribution function (CDF) of the standard bivariate normal distribution, Φ_2.

$$I_{11} = P(Z_1 > a, Z_2 > b) = P(Z_1 < -a, Z_2 < -b) = \Phi_2(-a, -b; \rho) = \Phi_2\left(\frac{\mu_1}{\sigma_1}, \frac{\mu_2}{\sigma_2}; \rho\right)$$

C
UNSUCCESSFUL ATTEMPTS TO CALCULATE I_{22}

Several approaches to calculate I_{22} proved to be dead ends.

C.1 Via the Identity on the Derivative of ϕ_2

Let's first prove the identity for the partial derivatives of the 2D normal PDF ϕ_2. For a bivariate normal distribution with correlation ρ :

$$\frac{\partial \phi_2}{\partial z_1} = -\frac{z_1 - \rho z_2}{1-\rho^2}\phi_2(z_1, z_2; \rho)$$
$$\frac{\partial \phi_2}{\partial z_2} = -\frac{z_2 - \rho z_1}{1-\rho^2}\phi_2(z_1, z_2; \rho)$$

Using these identities, we can write :

$$z_1\phi_2 = -\rho z_2 \phi_2 - (1-\rho^2)\frac{\partial \phi_2}{\partial z_1}$$

Integrating from a to ∞ with respect to z_1 :

$$\int_a^\infty z_1 \phi_2 \, dz_1 = -\rho z_2 \int_a^\infty \phi_2 \, dz_1 + (1-\rho^2)\phi_2(a, z_2; \rho)$$

Attempting to use this for I_{22}, we integrate $z_1 z_2 \phi_2$ over the region $[a,\infty) \times [b,\infty)$:

$$I_{22} = \int_b^\infty z_2 \left(\int_a^\infty z_1 \phi_2 \, dz_1 \right) dz_2$$

Substituting the identity :

$$I_{22} = -\rho \int_b^\infty z_2^2 \left(\int_a^\infty \phi_2 \, dz_1 \right) dz_2 + (1-\rho^2) \int_b^\infty z_2 \phi_2(a, z_2; \rho) \, dz_2$$

This approach becomes intractable because : 1. The first term involves a higher-order moment z_2^2 2. The second term requires evaluating another complex integral 3. Each attempt to simplify leads to terms of equal or greater complexity

Therefore, while the partial derivative identities are mathematically elegant, they do not provide a practical path to computing I_{22} when beta is non 0.

D
SPECIAL CASE : ZERO MEANS ($\mu_1 = \mu_2 = 0$)

In this case, the "from scratch" derivation leads to an elegant formula.

$$\mathbb{E}[\_\!/(X_1)\ \_\!/(X_2)] = \sigma_1\sigma_2 \int_0^\infty \int_0^\infty z_1 z_2 \phi_2(z_1, z_2; \rho)\, dz_1\, dz_2$$

The problem reduces to calculating $I_{22}(0,0;\rho) = \mathbb{E}[\_\!/(Z_1)\ \_\!/(Z_2)]$. The cleanest derivation uses Price's theorem :

1. $\frac{\partial \mathbb{E}[g]}{\partial \rho} = \mathbb{E}\left[\frac{\partial^2 g}{\partial z_1 \partial z_2}\right] = \mathbb{E}[\mathbb{1}(z_1 > 0, z_2 > 0)] = P(Z_1 > 0, Z_2 > 0) = \frac{1}{2\pi}\arccos(-\rho)$.
2. The integration constant is $\mathbb{E}[g]_{\rho=0} = \mathbb{E}[\_\!/(Z_1)]\mathbb{E}[\_\!/(Z_2)] = \left(\frac{1}{\sqrt{2\pi}}\right)^2 = \frac{1}{2\pi}$.
3. By integrating $\frac{\partial \mathbb{E}[g]}{\partial \rho}$ from 0 to ρ :

$$\begin{aligned}
\mathbb{E}[g](\rho) - \mathbb{E}[g](0) &= \int_0^\rho \frac{1}{2\pi}\arccos(-t)\, dt \\
&= \frac{1}{2\pi}\left[t\arccos(-t) + \sqrt{1-t^2}\right]_0^\rho \\
&= \frac{1}{2\pi}(\rho\arccos(-\rho) + \sqrt{1-\rho^2} - 1)
\end{aligned}$$

4. By adding the constant $\mathbb{E}[g](0) = 1/(2\pi)$, we get :

$$\mathbb{E}[\_\!/(Z_1)\ \_\!/(Z_2)] = \frac{1}{2\pi}(\sqrt{1-\rho^2} + \rho\arccos(-\rho))$$

The final formula for variances σ_1^2, σ_2^2 is :

$$\mathbb{E}[\_\!/(X_1)\ \_\!/(X_2)] = \frac{\sigma_1\sigma_2}{2\pi}(\sqrt{1-\rho^2} + \rho\arccos(-\rho))$$

D.1 Explicit Formulas for Σ and $\dot{\Sigma}$

For layer ℓ, the general formulas are :

$$\begin{aligned}
\Sigma^{(\ell)}(x,x') &= \mathbb{E}_{w,b}\left[\sigma(w^\top h^{(\ell-1)}(x) + \beta b)\sigma(w^\top h^{(\ell-1)}(x') + \beta b)\right] \\
\dot{\Sigma}^{(\ell)}(x,x') &= \mathbb{E}_{w,b}\left[\dot{\sigma}(w^\top h^{(\ell-1)}(x) + \beta b)\dot{\sigma}(w^\top h^{(\ell-1)}(x') + \beta b)w^\top w\right]
\end{aligned}$$

When $\beta = 0$, these simplify to :

$$\begin{aligned}
\Sigma^{(\ell)}(x,x') &= \mathbb{E}_{w}\left[\sigma(w^\top h^{(\ell-1)}(x))\sigma(w^\top h^{(\ell-1)}(x'))\right] \\
\dot{\Sigma}^{(\ell)}(x,x') &= \mathbb{E}_{w}\left[\dot{\sigma}(w^\top h^{(\ell-1)}(x))\dot{\sigma}(w^\top h^{(\ell-1)}(x'))w^\top w\right]
\end{aligned}$$

Note that for $\ell > 1$, both kernels become random fields due to the randomness in $h^{(\ell-1)}$.

D.2 Differential and Integral Form

The theorem relates the derivative of the expectation of a function g with respect to the covariance σ_{12} to the expectation of the second derivative of g. For $g = \mathrm{ReLU}(X_1)\,\mathrm{ReLU}(X_2)$, we have $\frac{\partial^2 g}{\partial x_1 \partial x_2} = \mathbb{1}(X_1 > 0, X_2 > 0)$.

$$\frac{\partial \mathbb{E}[\mathrm{ReLU}(X_1)\,\mathrm{ReLU}(X_2)]}{\partial \sigma_{12}} = \mathbb{E}[\mathbb{1}(X_1 > 0, X_2 > 0)] = I_{11}$$

This leads to the integral formulation :

$$\mathbb{E}[\mathrm{ReLU}(X_1)\,\mathrm{ReLU}(X_2)] = \int I_{11}(\sigma_{12})\, d\sigma_{12} + \text{Constant}$$

D.3 Calculating the Constant (Independent Case $\rho = 0$)

The constant is the value of the expectation when $\rho = 0$, i.e., when X_1 and X_2 are independent.

$$C = \mathbb{E}[\mathrm{ReLU}(X_1)] \cdot \mathbb{E}[\mathrm{ReLU}(X_2)]$$

The expectation of a single ReLU is :

$$\mathbb{E}[\mathrm{ReLU}(X)] = \sigma\phi_1\left(\frac{\mu}{\sigma}\right) + \mu\Phi_1\left(\frac{\mu}{\sigma}\right)$$

The constant is therefore the product of two such terms :

$$C = \left[\sigma_1\phi_1\left(\frac{\mu_1}{\sigma_1}\right) + \mu_1\Phi_1\left(\frac{\mu_1}{\sigma_1}\right)\right] \times \left[\sigma_2\phi_1\left(\frac{\mu_2}{\sigma_2}\right) + \mu_2\Phi_1\left(\frac{\mu_2}{\sigma_2}\right)\right]$$

This is the implementation of the constant in code, this is fairly long for non zeros beta, that's why we focus on the hypothesis with beta = 0. This in order to make experiments and proofs that MMNNs are superior to FCNNs for 2 hidden layer networks.

D.4 Exact moments and correlation for Kibble variables

Let $(X_i, Y_i)_{i=1}^r$ be i.i.d. Gaussian pairs with zero mean, unit variances and correlation ρ, and define

$$U = \sum_{i=1}^r X_i^2, \qquad V = \sum_{i=1}^r Y_i^2.$$

Then (U, V) follows Kibble's bivariate chi-square law with r degrees of freedom and correlation ρ. By Isserlis' (Wick's) theorem (or by expanding $\mathbb{E}[(X_i - \rho Y_i)^n (Y_i)^m]$ recursively) :

$$\mathbb{E}[X_i^2] = 1, \quad \mathrm{Var}(X_i^2) = 2, \quad \mathbb{E}[X_i^2 Y_i^2] = 1 + 2\rho^2, \quad (X_i^2, Y_i^2) = 2\rho^2.$$

Using independence across i :

$$\mathbb{E}[U] = r, \quad \mathrm{Var}(U) = 2r, \qquad \mathbb{E}[V] = r, \quad \mathrm{Var}(V) = 2r,$$

$$(U, V) = \sum_{i=1}^{r} (X_i^2, Y_i^2) = 2r\,\rho^2,$$

$$(U, V) = \frac{(U, V)}{\sqrt{\mathrm{Var}(U)\,\mathrm{Var}(V)}} = \frac{2r\rho^2}{\sqrt{(2r)(2r)}} = \rho^2,$$

$$\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] + (U, V) = r^2 + 2r\,\rho^2.$$

D.5 Product of Square Roots of Kibble Variables

Let $S = \sqrt{U}\sqrt{V} = \|X\|\|Y\|$. Due to correlation between U and V, we cannot directly multiply expectations. Instead, using Cauchy-Schwarz inequality :

$$\mathbb{E}[S^2] = \mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] + (U, V) = r^2 + 2r\rho^2$$

Therefore :

$$\mathbb{E}[S] \leq \sqrt{\mathbb{E}[S^2]} = r\sqrt{1 + \frac{2\rho^2}{r}}$$

The variance can be estimated similarly by expanding $(S - \mathbb{E}[S])^2$ and using correlation terms :

$$\mathrm{Var}(S) \approx \frac{r}{2}(1 + \rho^2) + O(1)$$

This suggests S/r approaches a distribution with mean $O(\sqrt{1 + \frac{2\rho^2}{r}})$ and variance of order $O(1/r)$.

D.6 Independence Properties and Asymptotic Behavior

The orientation variable r is independent of magnitudes (U, V). For the normalized product S/r, we have :

$$\frac{S}{r} = \frac{\|X\|\|Y\|}{r} \xrightarrow{r\to\infty} 1$$

This convergence occurs with fluctuations of order $O(1/\sqrt{r})$, showing that in high dimensions, the normalized product concentrates around 1 regardless of the correlation ρ.

E
EXPÉRIENCES NUMÉRIQUES

(paragraphe)